



**Course Code : CSE 3103**

**Course Name : System Analysis & Design**

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**Project Report: Agro Direct Supply Chain System**

**Submitted To :**

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## **Abstract :**

Often Farmers in the rural or semi-urban areas have to rely on several middlemen in order to directly access urban markets. These intermediaries raise prices for urban customers while decreasing farmers' profit margins. Significant post-harvest losses are also caused by ineffective delivery and transportation coordination. The AgroDirect System, a digital platform that directly links farmers with buyers. Farmers can market their produce, buyers can put direct orders or bids, logistics drivers can handle deliveries, and administrators can keep an eye on system activity. The AgroDirect System seeks to boost farmers' income, lower consumer costs, and limit post-harvest losses by doing away with needless middlemen, guaranteeing transparent pricing, and enhancing logistics management.

## **Introduction:**

### **Problem Description:**

Currently, farmers in rural areas lack direct access to urban markets. They rely on multiple layers of middlemen who suppress buying prices, while urban consumers pay high prices due to markups. Additionally, post-harvest loss is high due to poor logistics coordination.

## **Purpose :**

The primary purpose of this document is to formally introduce the AgroDirect Supply Chain System, serving as a comprehensive blueprint for its design, development, and implementation. It establishes a shared understanding of the system's requirements, architectural configuration, and intended functionality among all stakeholders, ensuring that the final solution aligns with the established business needs.

## **Document Scope:**

This document covers the complete system analysis and design phases of the AgroDirect project, including:

- **Requirements:** Defining the capabilities (Use Cases) and data flow (DFD).
- **Design:** Presenting the technical structure (System Architecture), database blueprint (ER and Class Diagrams), and process logic (Sequence and Activity Diagrams).
- **Timeline:** Providing a planned schedule for project completion (Gantt Chart).

This document does not cover the specific user interface (UI) wireframes or low-level coding details.

## System Overview:

The AgroDirect System is an online platform designed to facilitate the direct exchange of fresh agricultural produce between Farmers and Buyers by eliminating traditional middlemen.

- **Core Functionality:** It enables Farmers to upload inventory, allows Buyers to place direct orders or participate in bidding, and manages the entire logistics chain through integrated Drivers and a secure Payment Gateway.
- **Architectural Style:** The system is built on a robust server farm architecture featuring a Load Balancer, Application Server, and a PostgreSQL Database for reliability and scalability.

## Business Context:

The AgroDirect System addresses key challenges in the current agricultural supply chain:

- **Value for Farmers:** Ensures Farmers receive fairer compensation by directly connecting them to the market.
- **Value for Buyers:** Provides consumers with fresher produce at competitive prices due to a shortened supply chain.
- **Efficiency:** Streamlines logistics and payment processing, introducing transparency and speed into the distribution process. The goal is to complete the project within the defined timeline, concluding with deployment and monitoring by May 2026.

## Stakeholders:

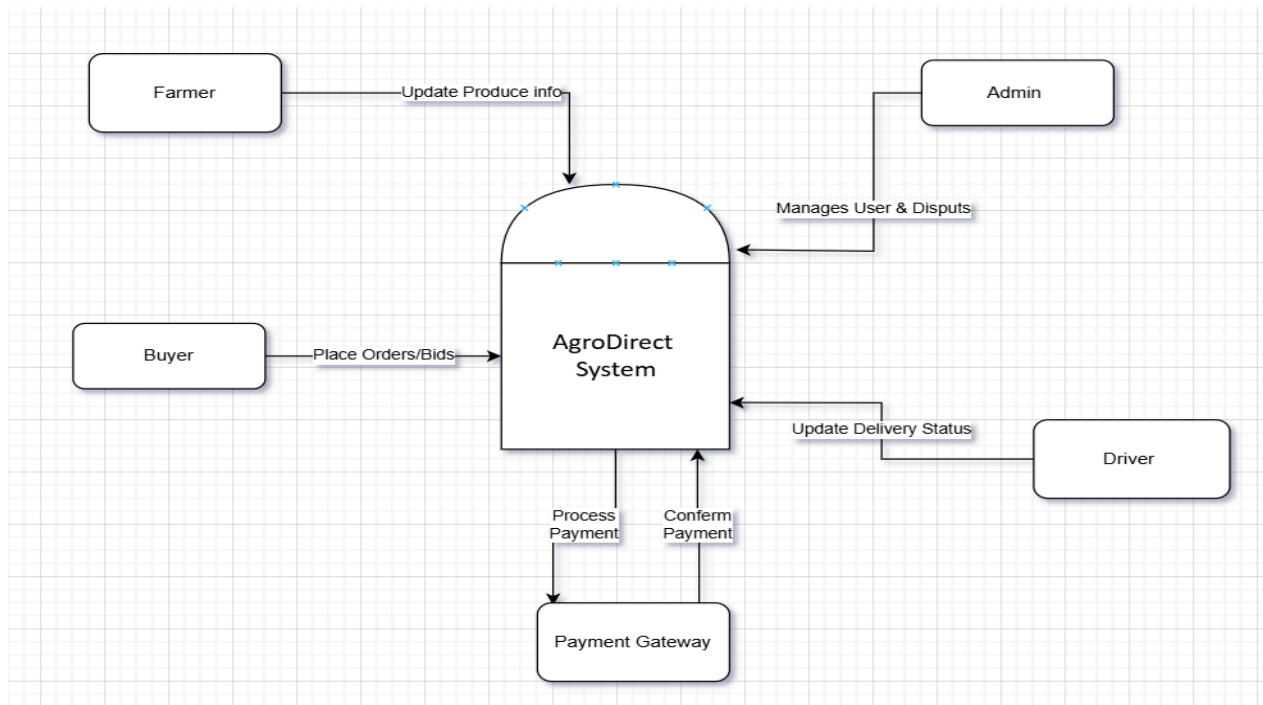
The following key groups and individuals have a vested interest in the success of the AgroDirect System:

- **Primary Users (Actors):**
  - **Farmers:** Upload and manage crop inventory, and accept orders.
  - **Buyers:** Search products, place orders, and make payments.
  - **Drivers (Logistics):** Receive delivery jobs and update shipment status.
- **System Managers:**
  - **Admin:** Manages user registration, monitors system activity, and views sales reports.

- **Project Team:**
  - System Analysts, Designers, Developers, and Testers responsible for the project's execution.

## System Diagrams:

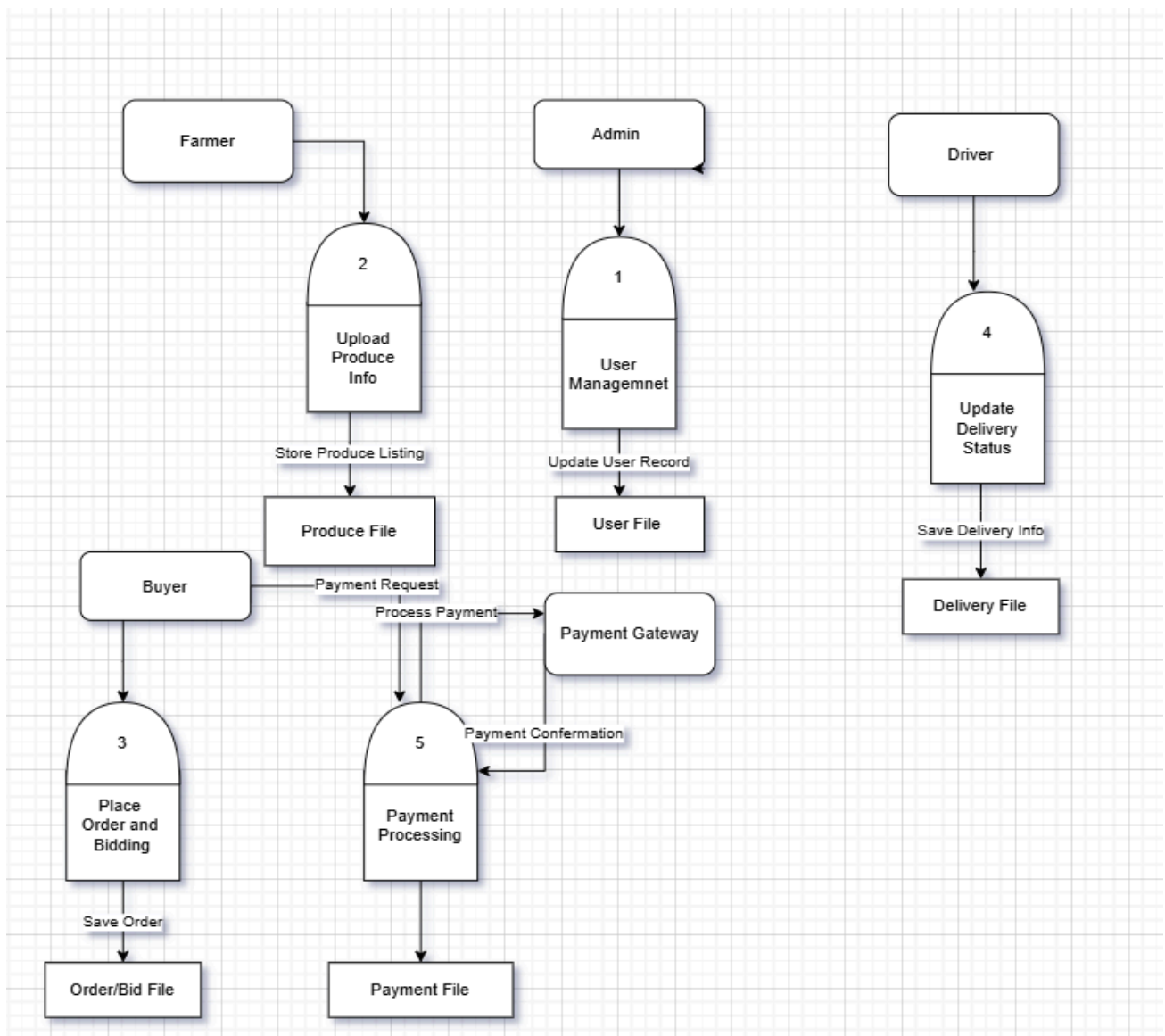
### A. Context Diagram :



### Description :

This provided activity diagram describes how different users interact with the AgroDirect System and how the system communicates with external services. The four primary participants in the process are the farmer, the buyer (retailer/consumer), the logistics driver, and the administrator. Produce details, including crop kind, quantity, and price, are uploaded into the AgroDirect System by the farmer. To buy the produce, the buyer uses the system to place orders or submit bids. During transportation, the logistics driver modifies the delivery status (picked up, in transit, delivered) in the system. The administrator controls all system functions, manages users, and resolves conflicts. The AgroDirect System communicates with the Payment Gateway within the system to handle payments after an order is placed. The gateway returns a payment confirmation to the system following payment processing.

## B. DFD -0 Diagram:



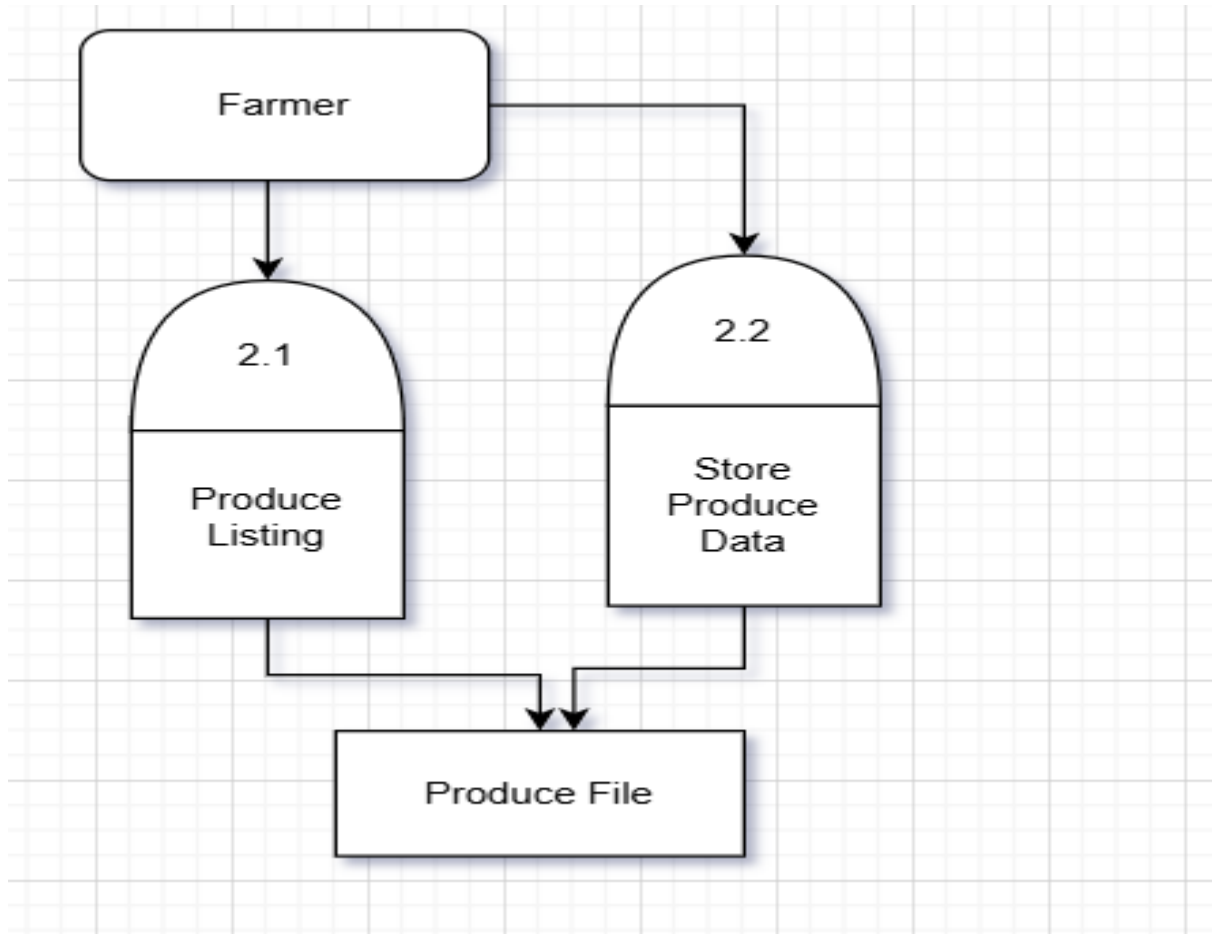
This diagram maps out exactly how information flows through the entire system. It shows the four main users: the Farmer, Buyer, Admin, and Driver. The Farmer uploads produce details, while the Buyer places orders and uses the Payment Gateway for transactions. The Admin manages all user records, and the Driver updates delivery status. All crucial information is saved in specific files, linking every process and user together.

## DFD-1 :

### 2. Upload produce Info

#### 2.1: Producing Listing

#### 2.2: Store Produce Data

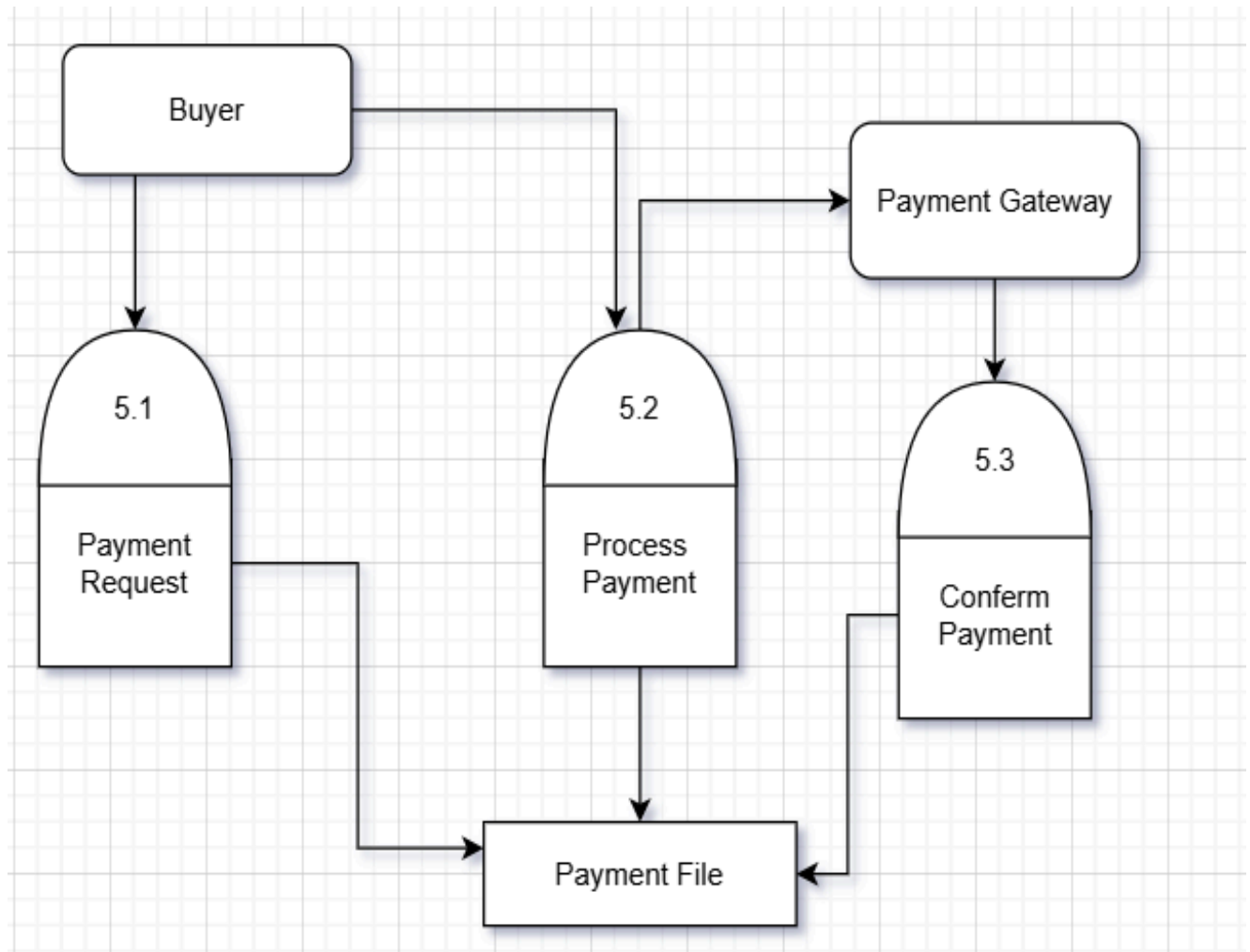


## 5. payment processing

### 5.1:Payment Request

### 5.2:Process Payment

### 5.3:Confirm Payment



## **1.User Management**

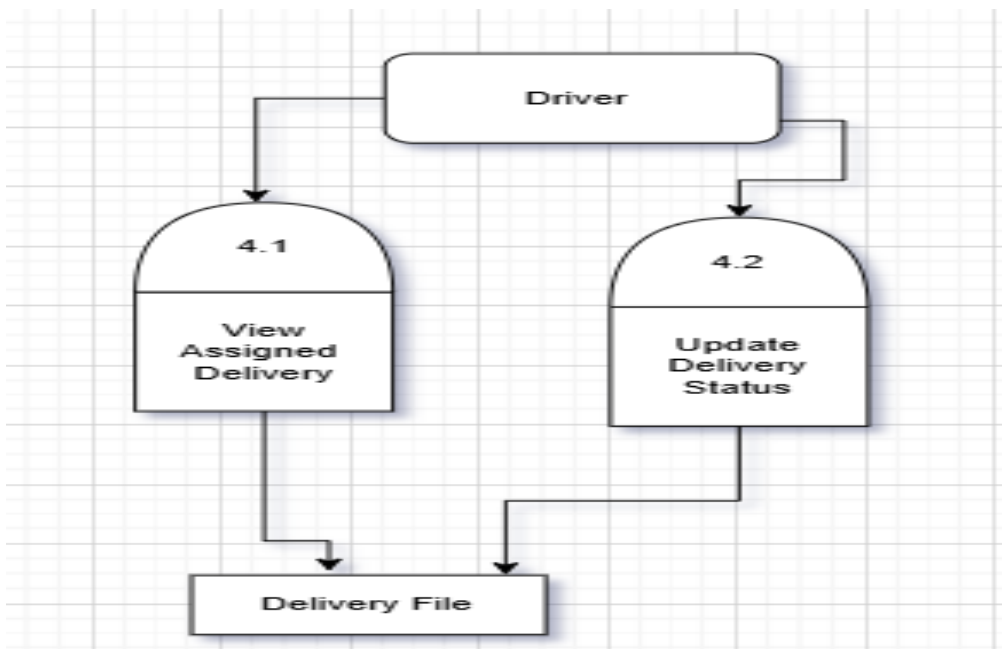
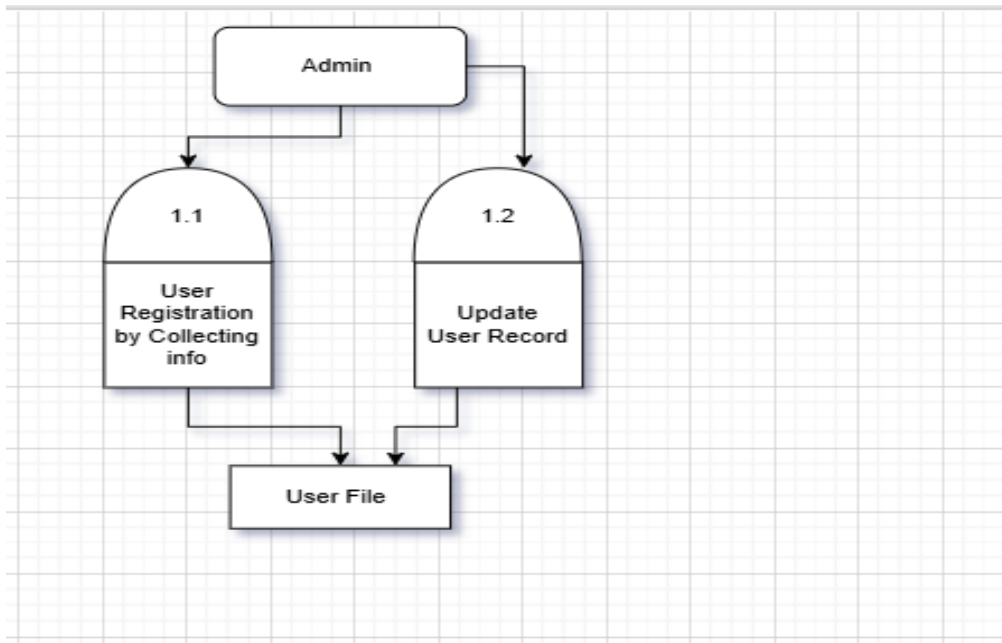
**1.1:User Registration by collecting info**

**1.2:Update User Record**

**4.update delivery status**

**4.1 view assigned delivery**

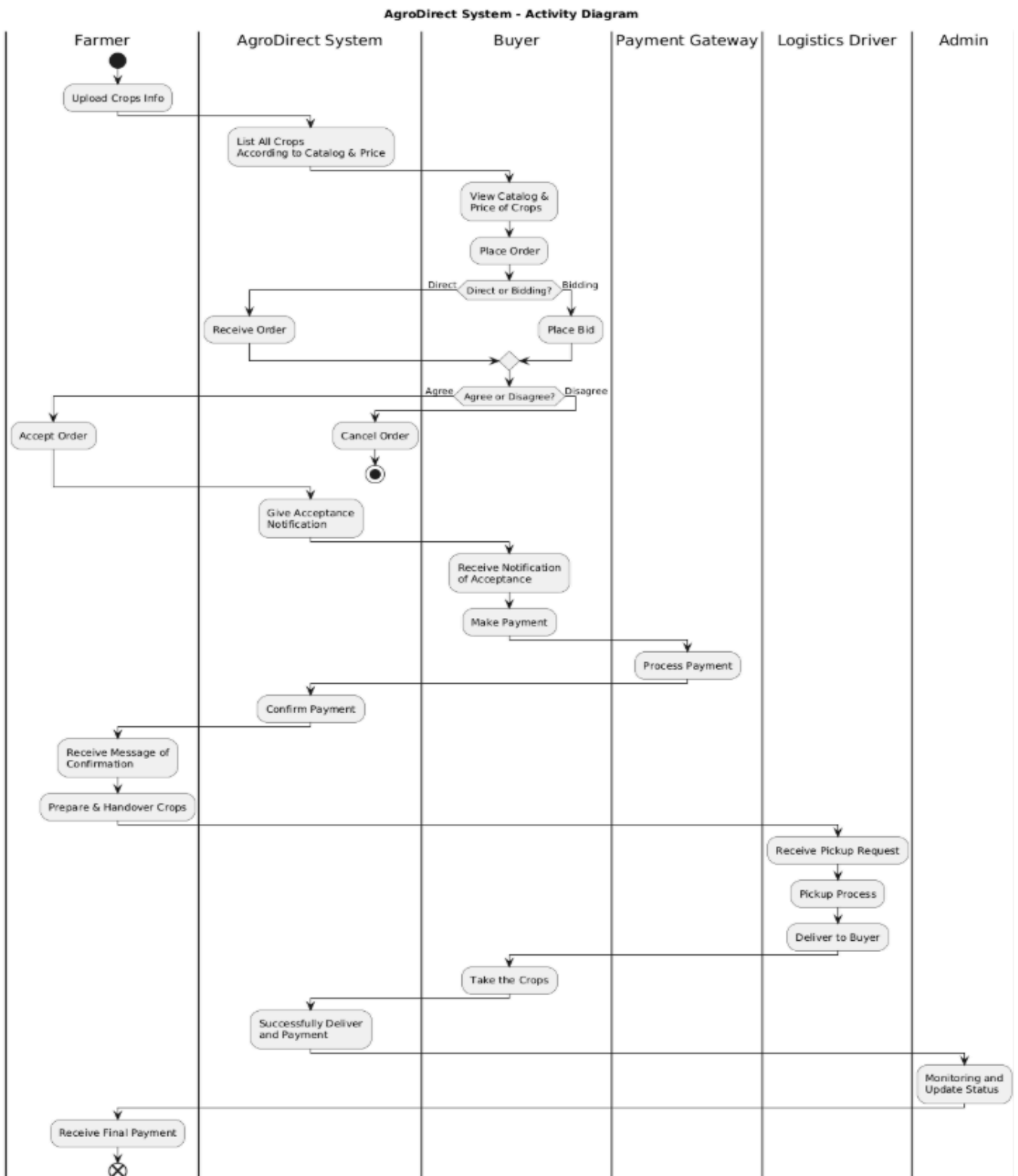
**4.2 update delivery status**



These Level 1 DFD diagrams provide a detailed breakdown of all the major functions within the Agro Direct Supply Chain System. The Admin manages all user records, while the Farmer handles the upload of produce listings. The Buyer places orders, participates in bidding, and completes payments through the Payment Gateway. Finally, the Driver updates the delivery status to ensure transparency in the supply chain. All information is meticulously organized and stored in dedicated Files such as the User File, Produce File, and Payment File.

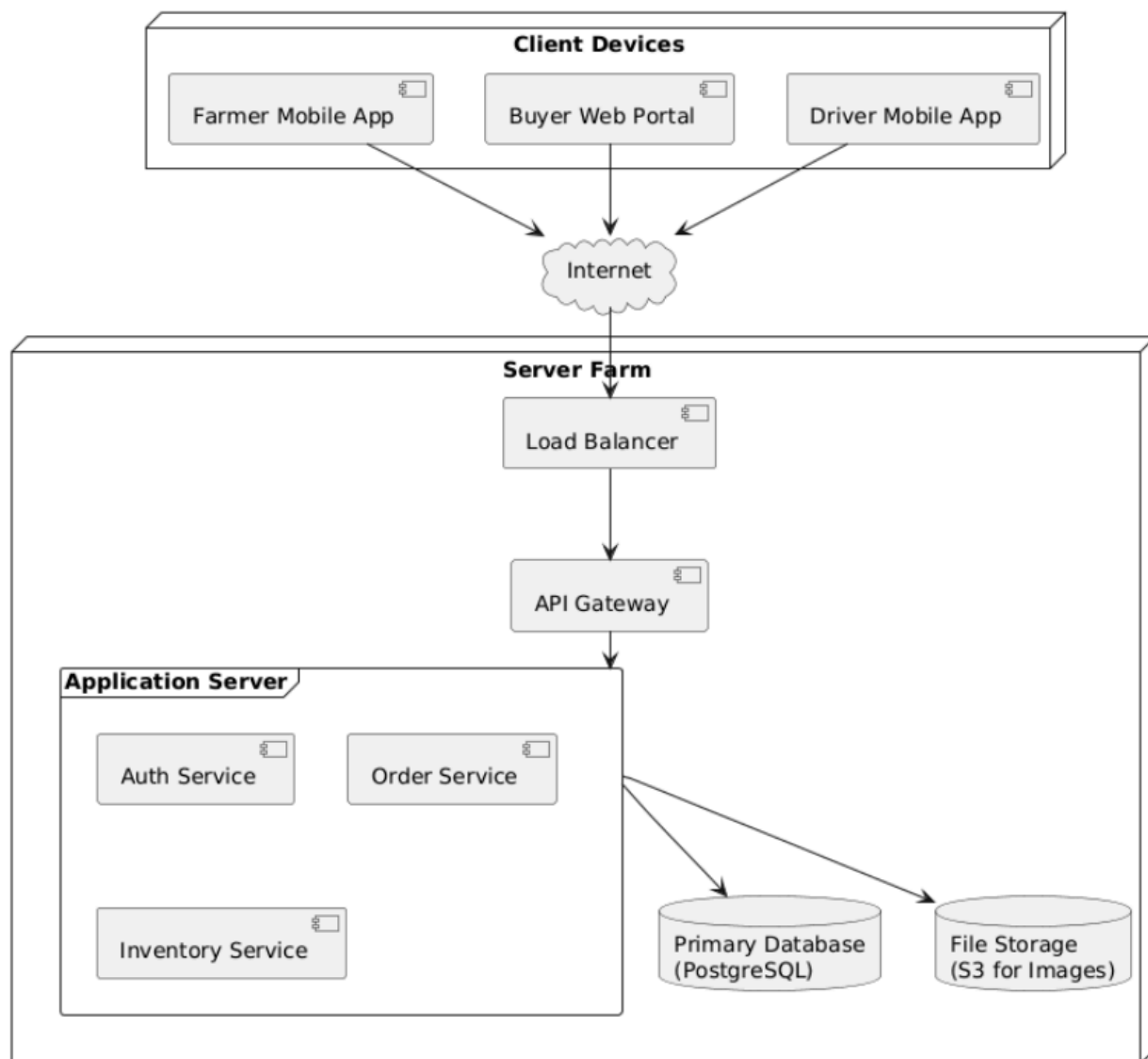


## C. Activity Diagram:



This diagram maps out the complete flow of actions and decisions that take place across the entire system. It starts with the Farmer uploading crop information and the Buyer viewing the catalog and placing an order, which includes an option for bidding. After the Farmer agrees to the bid or direct order, the Buyer makes a payment which is handled by the Payment Gateway. Once payment is confirmed, the Logistics Driver receives a pickup request and delivers the crops, while the Admin monitors the whole process. The flow ends when the Farmer receives the final payment.

#### D. System Architectural Configuration:



This diagram shows the technical structure of the entire system, detailing how users connect to the server. The Farmer, Buyer, and Driver use different devices (like mobile apps and web portals) to connect through the Internet to the Server Farm. The Server uses a Load Balancer and API Gateway to manage incoming user requests. These requests are handled by the Application Server, which contains the core functions (like Order and Authentication Services). Finally, all important information is stored in the Primary Database (PostgreSQL) and the separate File Storage for items like images.

## **Functional & Non-Functional Requirements:**

### **Functional Requirements**

1. **User Management:** Registration/Login for Farmers, Buyers, and Drivers.
2. **Inventory Management:** Farmers can upload crop details (photos, quantity, harvest date).
3. **Order Processing:** Buyers can add items to cart and checkout.
4. **Bidding System:** Buyers can place bids on bulk quantities.
5. **Logistics Matching:** Drivers receive notifications for nearby pickup requests.
6. **Payment Integration:** Secure handling of transactions.
7. **Rating System:** Users can rate farmers and drivers.

### **Non-Functional Requirements**

1. **Scalability:** Must handle concurrent users during harvest season.
2. **Usability:** Mobile-first design for farmers with simple UI/UX.
3. **Security:** End-to-end encryption for payment data.
4. **Availability:** 99.9% uptime required.

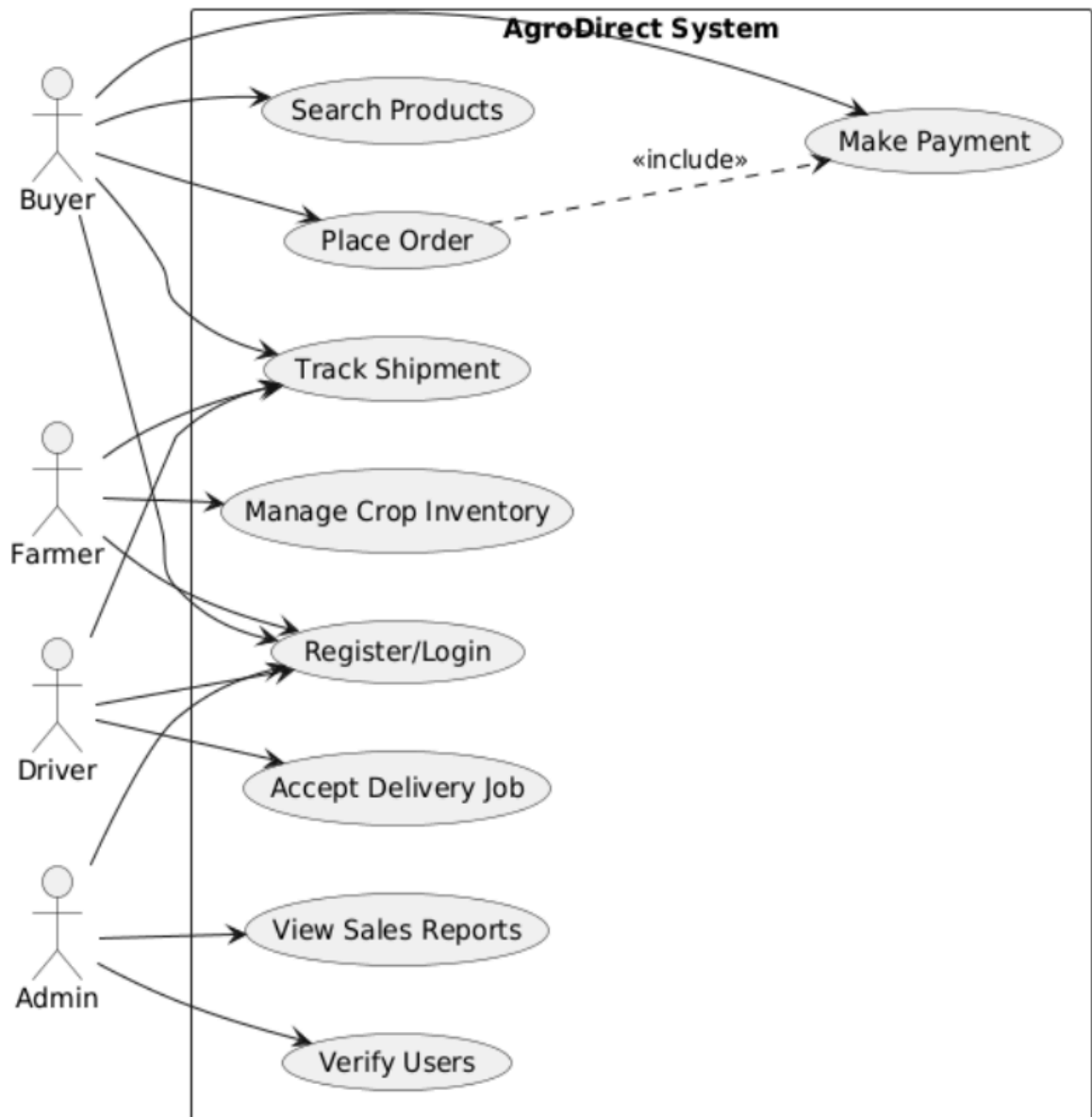
**Requirements Description:**

| Identifier | Requirement Description | PW |
|------------|-------------------------|----|
| REQ1       | User Sign up            | 5  |
| REQ2       | Upload Crop Info        | 3  |
| REQ3       | Search/Filter Crops     | 4  |
| REQ4       | GPS Tracking            | 2  |

**Traceability Matrix (Sample)**

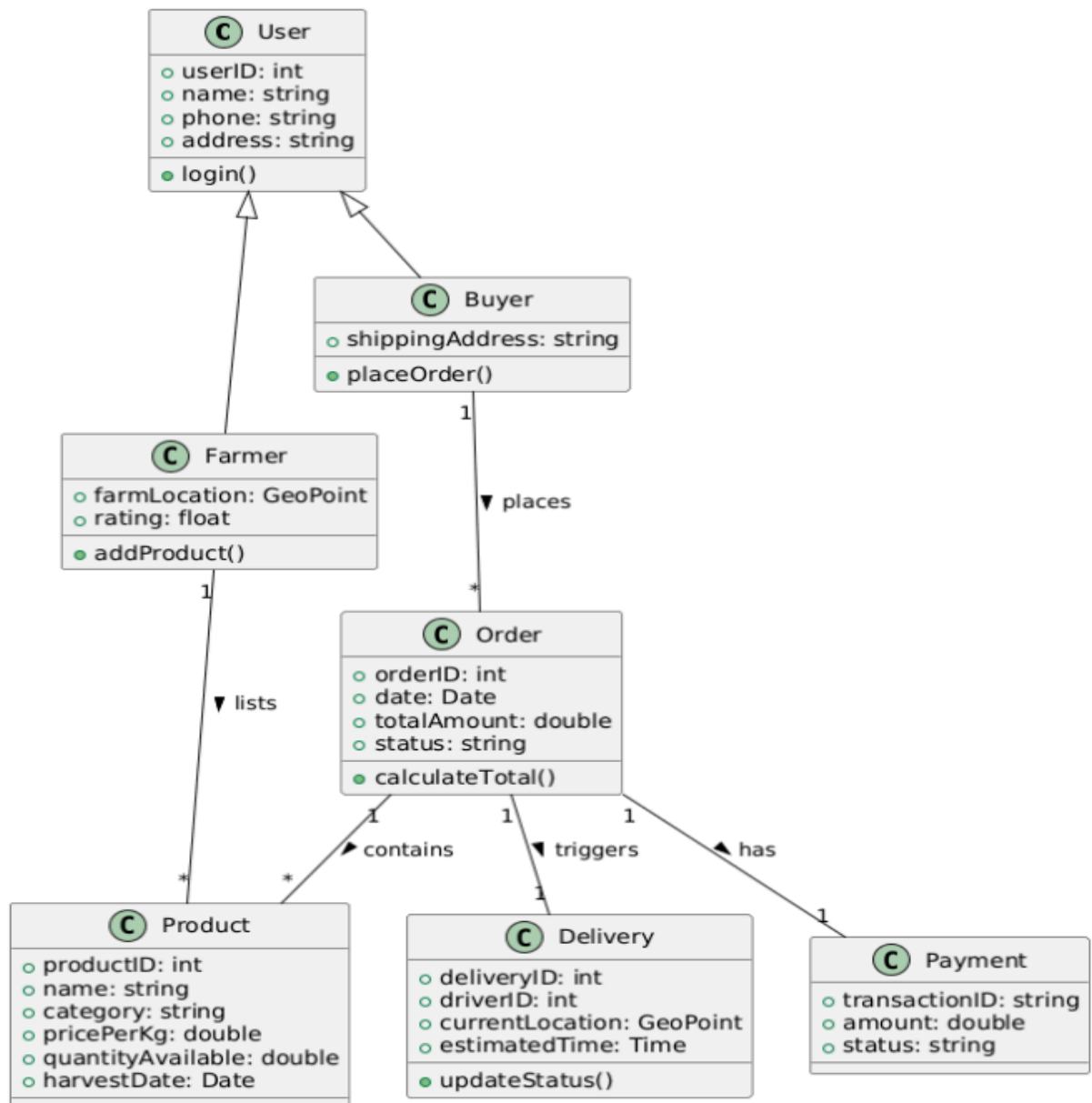
| Req ID | Description         | UC1 (Reg) | UC2 (Post Crop) | UC3 (Order) | UC4 (Delivery) |
|--------|---------------------|-----------|-----------------|-------------|----------------|
| REQ1   | User Sign up        | X         |                 |             |                |
| REQ2   | Upload Crop Info    |           | X               |             |                |
| REQ3   | Search/Filter Crops |           |                 | X           |                |
| REQ4   | GPS Tracking        |           |                 |             | X              |

## E. Use Case Diagram:



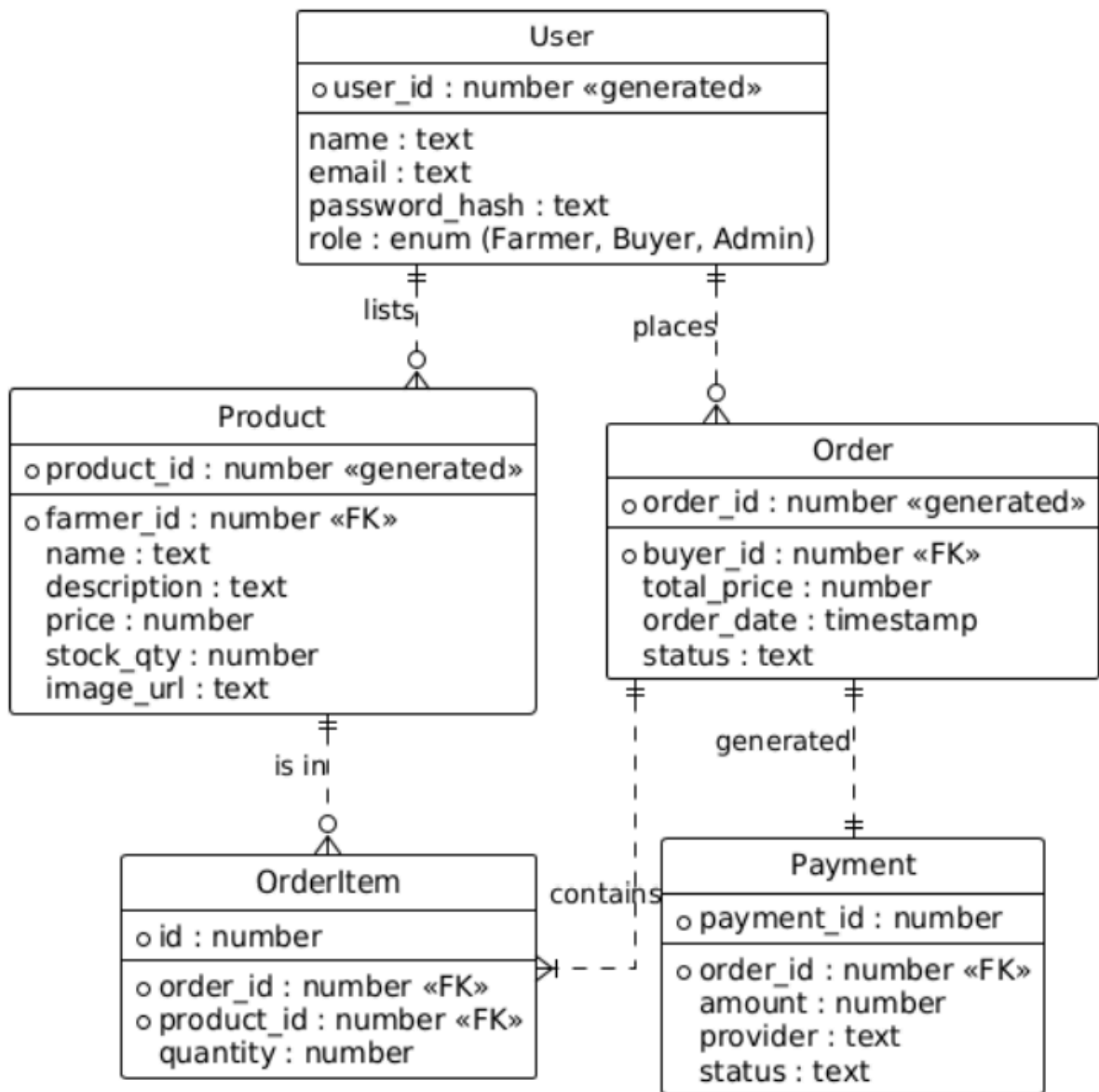
This diagram outlines all the main functions available within the AgroDirect System. The Buyer, Farmer, Driver, and Admin are the key users (Actors), and the diagram shows what specific tasks each can perform. For example, the Buyer can 'Search Products' and 'Place Order', which always includes the 'Make Payment' step. The Farmer can 'Manage Crop Inventory', the Driver can 'Accept Delivery Job', and the Admin can 'Verify Users' and 'View Sales Reports'.

## F. Domain Level Class Diagram:



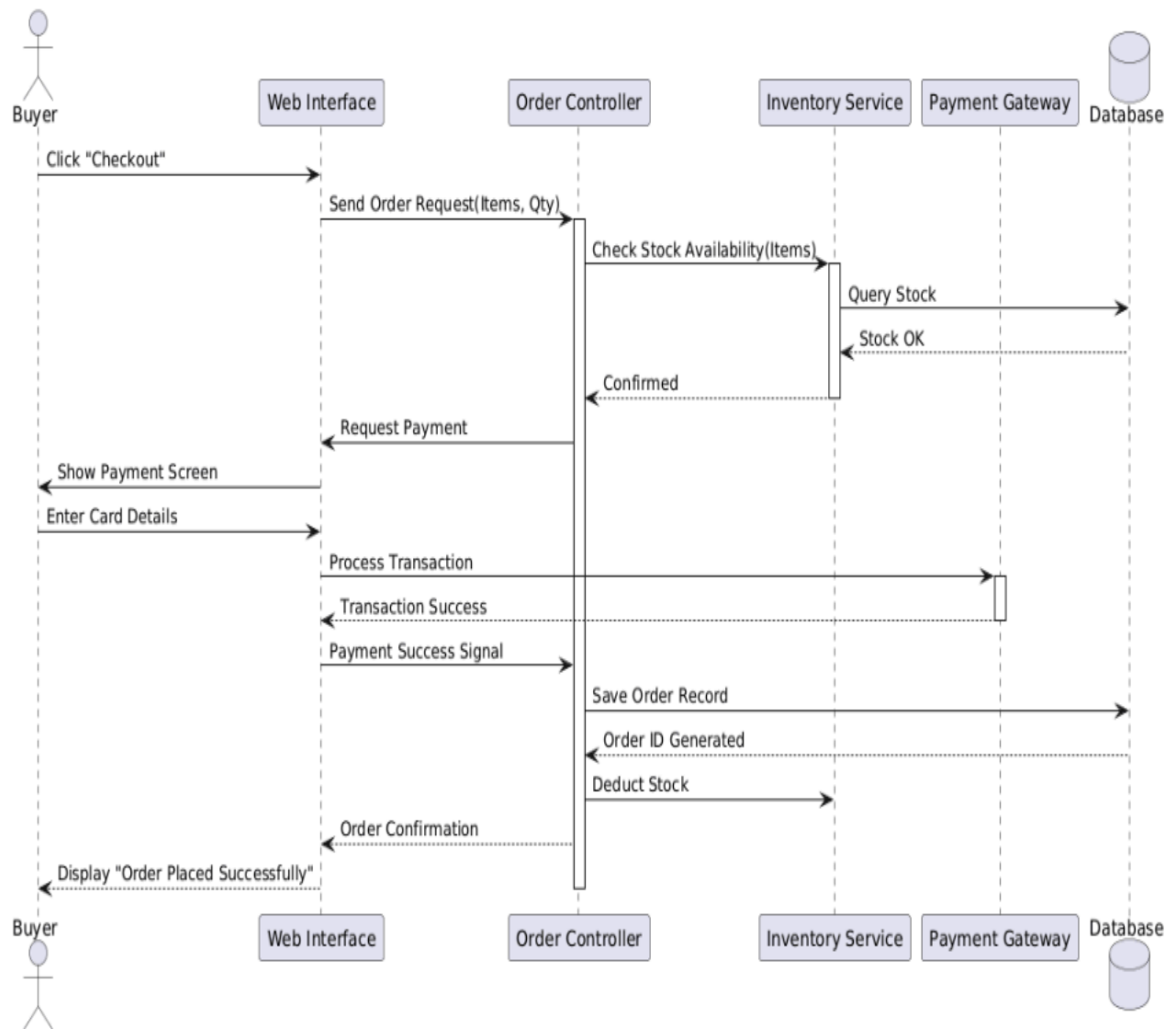
This diagram shows the main objects (or *classes*) in the system and how they are all connected. It defines the key users—the Farmer and the Buyer—who are special types of a general User. It shows that a Farmer lists a Product, and a Buyer places an Order. An Order *contains* Products and triggers a Delivery. Each Order also has a Payment, which keeps track of the money transaction. This structure guides how the system saves and manages all its data.

## G. ER Diagram (Entity Relationship):



This diagram shows the database structure, defining the main tables (Entities) and how they are related using keys. It shows that the main User table is linked to Product (through the Farmer role) and Order (through the Buyer role). An Order consists of multiple OrderItem records, which link back to the specific Product sold. Crucially, every Order generates a Payment record, ensuring financial tracking. This diagram uses concepts like Primary Keys (generated ID) and Foreign Keys (FK) to show how data is connected across the tables.

## H. Sequence Diagram (Place Order Process):



This diagram shows the exact order of steps (or *sequence*) that happens when a Buyer places an order. The Buyer clicks "Checkout" which sends a request to the Order Controller. The Controller first checks the Inventory Service (database) to confirm stock. If available, the system requests payment through the Payment Gateway. After a successful transaction, the Controller saves the new Order record in the Database, reduces the stock quantity, and finally sends a confirmation back to the Buyer.

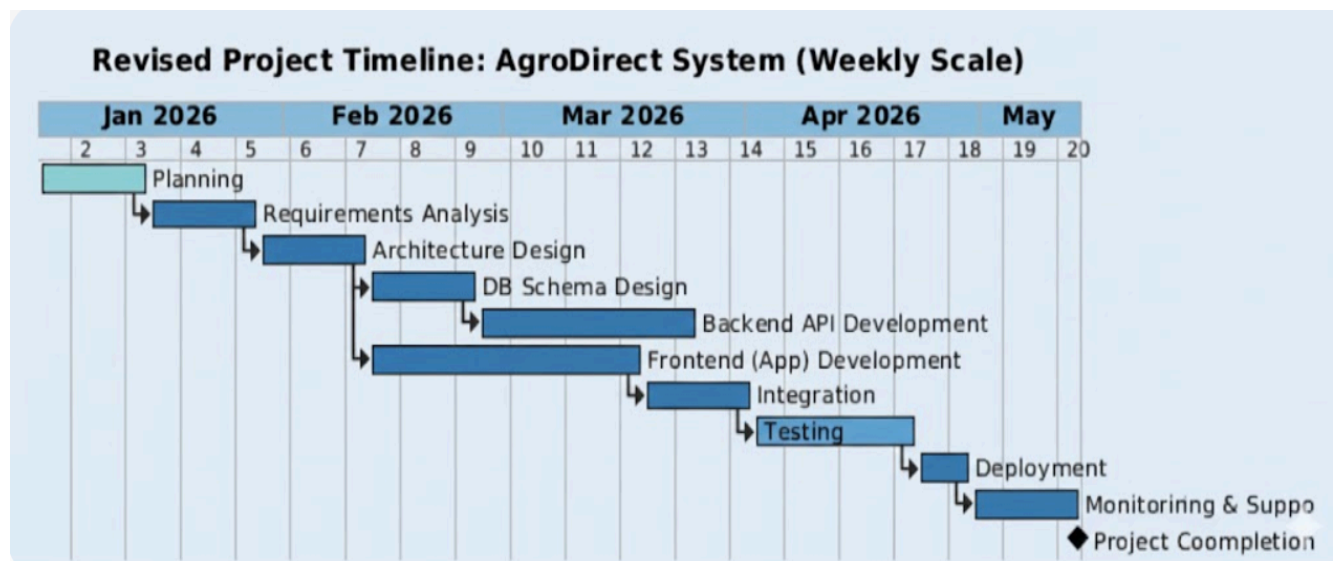


## Project Plan & Cost Estimation :

### Task -Duration -Prerequisite

- 1.Planning -2 weeks — (Project Initiation)
- 2.Requirements Analysis - 2 weeks -(Planning)
- 3.Architecture Design- 2 weeks -(Requirements Analysis)
- 4 DB Schema Design 2 weeks-( Architecture Design)
- 5 Backend API Development- 4 weeks- (DB Schema Design, Architecture Design )
- 6.Frontend (App) Development -5 weeks -(Backend API Development, Architecture Design )
- 7.Integration -2 weeks- (Backend API Development, Frontend (App) Development)
- 8.Testing -3 weeks -(Integration, Backend API Development)
- 9.Development- 1 weeks-(Testing)
- 10.Monitoring & Support - 3 weeks -(Development)

### Grantt Chart :

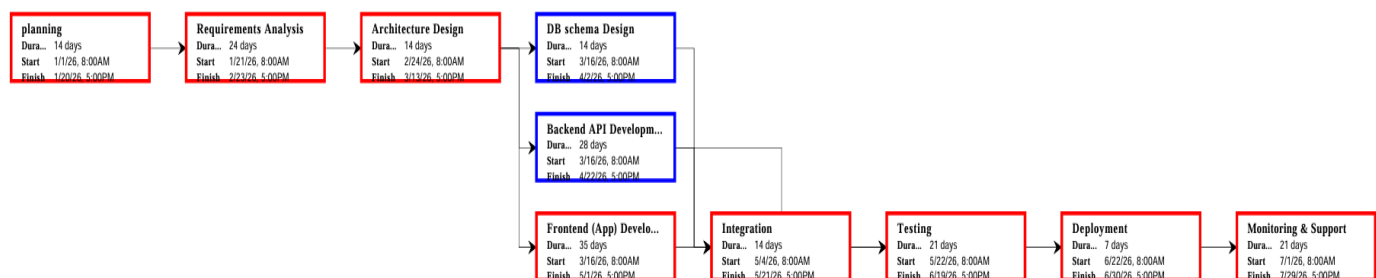


This chart shows the complete plan and schedule for the AgroDirect System project, organized over a weekly scale from January to May 2026. It starts with Planning and Requirements Analysis in January. The core development tasks like Architecture Design, DB Schema Design, Backend API Development, and Frontend (App) Development take place from late January through March. The final stages involve Integration and Testing in April, followed by Deployment and Monitoring & Support in May, leading to the Project Completion.

## Network Diagram:

| SI | Description             | Predecessor | Days |
|----|-------------------------|-------------|------|
| 1  | Planning                | -----       | 14   |
| 2  | Requirements Analysis   | 1           | 14   |
| 3  | Architecture Design     | 2           | 14   |
| 4  | DB Schema Design        | 3           | 14   |
| 5  | Backend API Development | 3           | 28   |
| 6  | Frontend Development    | 3           | 35   |
| 7  | Integration             | 4,5,6       | 14   |
| 8  | Testing                 | 7           | 21   |
| 9  | Deployment              | 8           | 7    |
| 10 | Monitoring & Support    | 9           | 21   |

## Diagram with critical path:



## Total Cost :

### Hardware:

| Hardware name | Quantity | Cost(BDT)                      |
|---------------|----------|--------------------------------|
| Computer      | 5        | $100,000 \times 10 = 1000,000$ |
| printer       | 3        | $20,000 \times 3 = 60,000$     |

### Software:

| Software name     | Time     | Cost    |
|-------------------|----------|---------|
| Database server   | 8 months | 400,000 |
| Domoin C Hosting  | 8 months | 20,000  |
| API subscription  | 8 months | 10,000  |
| DB server testing | 2 months | 200,000 |

### Human Resource:

| Human Resource Role | Number of Human Resource | Cost per month                 |
|---------------------|--------------------------|--------------------------------|
| System Analyst      | 1                        | $(120,000 \times 1) = 120,000$ |
| Sysyem designer     | 1                        | $(150,000 \times 1) = 150,000$ |
| Programmer          | 6                        | $(80,000 \times 6) = 480,000$  |
| System administator | 1                        | $(100,000 \times 1) = 100,000$ |
| System manager      | 1                        | $(90,000 \times 1) = 90,000$   |
| Office Staff        | 4                        | $(25,000 \times 4) = 100,000$  |

**Utility:**

| Name             | Time     | Cost    |
|------------------|----------|---------|
| Office rent      | 8 months | 400,000 |
| Utility bill     | 8 months | 120,000 |
| Maintenance cost | 8 months | 200,000 |

**Benefit estimation:**

| Benefit Type         | Description                                       | Estimated Annual Value(BDT) |
|----------------------|---------------------------------------------------|-----------------------------|
| Subscription Revenue | Monthly subscription from user                    | 1,200,000                   |
| Ad Revenue           | Revenue from ads displayed in the free version    | 300,000                     |
| API Licensing        | Licensing the forecast API to third parties       | 600,000                     |
| Total Tangible       |                                                   | 2,100,000                   |
| User Trust           | Improve public trust in Weather alerts            | High                        |
| Brand Value          | Enhanced reputation as a reliable weather service | High                        |
| Data Accuracy        | More reliable forecast for agriculture            | Significant                 |

## Break even point analysis:

| Year | Cumulative cost | Annual benefit | Cumulative benefit | Net Cash Flow |
|------|-----------------|----------------|--------------------|---------------|
| 0    | 3,350,000       | 0              | 0                  | -3,350,000    |
| 1    | 3,850,000       | 15,00000       | 15,00000           | -23,50000     |
| 2    | 4,250,000       | 20,00000       | 35,00000           | -7,50000      |
| 3    | 4,650,000       | 30,00000       | 65,00000           | 18,50000      |

Break-Even Point Calculation:

Using the net Cash flow values:

Year1: Still negative (-23,50000 BDT)

Year2: negative (-7,50000 BDT)

Year3: positive(+18,50000)

**Break even occurs between year 2 and year 3**

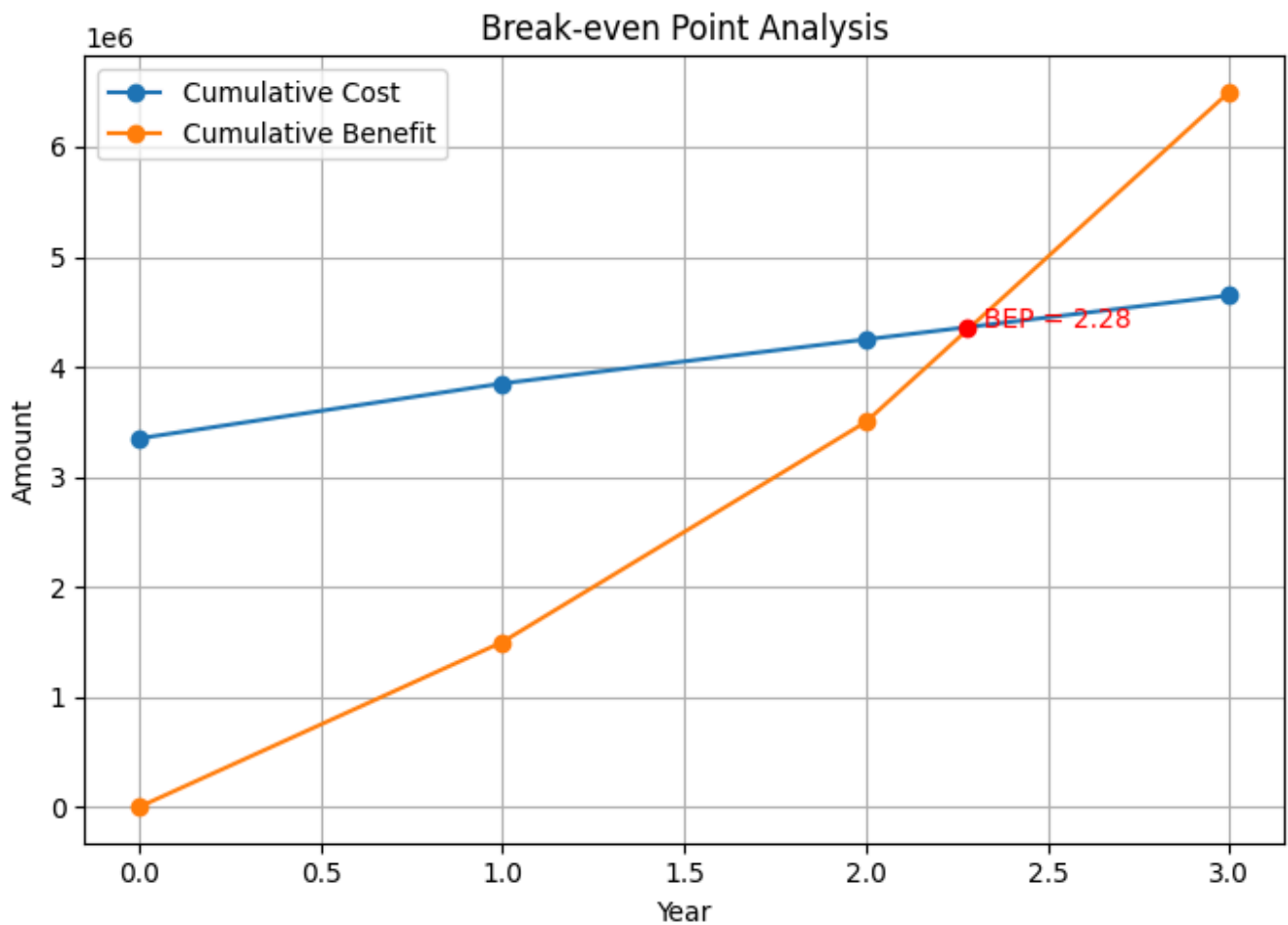
$$\text{BEP (year)} = \text{Year before positive NCF} + (\text{NCF at Year before}) / (\text{NCF at Year after} - \text{NCF at Year before})$$

NCF at Year 2 = -7,50,000

NCF at Year 3 = 18,50,000

$$\text{BEP} = 2 + |-7,50,000| / 18,50,000 - (-7,50,000) = 2.28 \text{ years}$$

## Graph:



## Conclusion :

The AgroDirect System offers a complete solution to the major problems that rural farmers and urban customers have to face in the conventional agricultural supply chain. The approach ensures fair pricing and increased profitability for farmers by removing needless middlemen and facilitating direct communication between farmers and buyers. Coordinated logistics management minimizes post-harvest losses and delivery delays, while integrated payment processing improves transaction security. System dependability and conflict resolution are guaranteed by administrative supervision. All things considered, the AgroDirect System encourages sustainability, efficiency, and transparency in agricultural trading and has the potential to greatly strengthen the connection between rural and urban markets.